MUTUAL FUND STYLE DRIFT & ANOMALY DETECTION

A Complete Theory Primer — From Zero to Project-Ready

Scope of this document

Pure theory — no code, no implementation steps.
Covers mutual fund fundamentals, SEBI/AMFI regulatory structure,
financial metrics, and the ML concepts needed for a
Mutual Fund Style Drift Detection project.

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How to Use This Document

This is a theory-only reference. It will not show you how to write a single line of code. Its only job is to make sure that by the time you sit down to build the project, you already understand every financial concept, every regulation, and every ML concept well enough to explain it out loud — in an interview, to a friend, or to yourself.

The document is organised into three layers, matching how the topic actually builds on itself:

The Three Layers

Layer 1 — Mutual Fund Fundamentals: what a fund is, how it is structured, and why categories exist.

Layer 2 — Regulatory Framework & Financial Metrics: the exact SEBI/AMFI rules and the numeric tools used to measure a fund's behaviour.

Layer 3 — The ML & Statistical Concepts: clustering, anomaly detection, dimensionality reduction, and time series reasoning — explained purely conceptually, mapped back to this specific project.

Each topic follows the same structure: a plain-English explanation, why it matters specifically for style drift detection, and a short list of trusted resources (articles, official sources, and YouTube channels) to go deeper.

LAYER 1

Mutual Fund Fundamentals

Everything in this layer answers one question: what exactly is the "thing" your project is analysing? Before you can detect a fund behaving abnormally, you need a rock-solid mental model of what normal behaviour even means.

1.1 What Is a Mutual Fund, Really?

A mutual fund is a pooled investment vehicle. Thousands of individual investors each contribute money into a common pool. A professional fund manager then invests that pooled money into a basket of securities — stocks, bonds, or both — according to a stated objective.

Each investor owns "units" of the fund, similar to owning shares of a company. The value of one unit is called the Net Asset Value (NAV), and it is calculated once at the end of every trading day.

Why this matters for your project:

Your entire dataset is built from the fact that every mutual fund must legally disclose exactly which stocks it holds, in what quantity, every single month. That disclosure — the portfolio holding statement — is the raw material your clustering model will run on.

Key terms to know cold

Term	Plain-English Meaning
AMC (Asset Management Company)	The company that legally operates the mutual fund (e.g. HDFC AMC, SBI Mutual Fund).
Fund Manager	The individual or team that decides which stocks to buy/sell inside the fund.
NAV (Net Asset Value)	The price of one unit of the fund, calculated daily as (Total Assets – Liabilities) ÷ Total Units.
Scheme	The specific fund product, e.g. "HDFC Large Cap Fund."
AUM (Assets Under Management)	The total money currently invested in the fund. A growing AUM means more inflows; the fund manager must deploy more capital.
Expense Ratio	Annual fee charged by the AMC, expressed as a % of AUM, to manage the fund.
Benchmark Index	The index a fund compares itself against (e.g. Nifty 100 for large cap funds) to judge if it is outperforming or underperforming.

1.2 Types of Plans Within a Fund

These distinctions don't directly feed your clustering model, but you need them to correctly read any fund factsheet or portfolio disclosure without confusion.

Direct Plan vs. Regular Plan

A Direct Plan is bought straight from the AMC with no distributor commission, so its expense ratio is lower and returns are slightly higher. A Regular Plan is bought through an intermediary (broker, advisor) who earns a trail commission, baked into a higher expense ratio. Both plans of the same fund hold an identical portfolio — only the cost and resulting NAV differ.

Growth Plan vs. Dividend (IDCW) Plan

In a Growth Plan, profits stay invested and compound inside the fund — the NAV simply rises. In a Dividend / IDCW (Income Distribution cum Capital Withdrawal) Plan, the fund periodically pays out a portion of profits to investors, and the NAV drops by that paid-out amount. The underlying portfolio is identical in both cases.

Think of it like an apple tree (edited-for understanding)

- In a **Growth Plan**, you leave the apples on the tree to grow bigger and multiply.
- In an **IDCW Plan**, you periodically pick some apples (take the cash) and keep them, meaning fewer apples remain on the tree.

Active vs. Passive Funds

An actively managed fund has a human fund manager picking stocks, trying to beat a benchmark. A passive fund (index fund / ETF) simply replicates a benchmark index exactly, with no manager discretion. Style drift, by definition, is only a meaningful concern for actively managed funds — a passive index fund cannot drift because it is mechanically forced to mirror its index.

The Fruit Salad Analogy (edited example for understanding)

Think of building a fruit salad by buying pre-mixed baskets:

- **The Passive Index Fund (The Fruit Basket):** This basket is assembled by an automated machine to perfectly match a standard "Mixed Fruit" recipe. If the recipe calls for exactly (40%) apples, (40%) oranges, and (20%) grapes, the basket will mirror this exactly. The machine has no ability to change the mix.
- **The Actively Managed Fund (The Personal Chef):** You hire a chef (the fund manager) and ask them to make you a (100%) Fruit and Vegan salad. They happily agree and start with a great fruit mix.

What is Style Drift for this example?

After a few months, you check the chef's salad bowl and notice they have added a few slices of cheese and chunks of bacon.

When you ask why, the chef explains: *"Fruit prices are up, and I thought adding bacon and cheese would make the salad taste better and earn higher praise from food critics."*

Even though the chef's modifications might make the salad taste great, **they drifted from your original request** a (100%) fruit salad). You now have an entirely different meal than you signed up for.

1.3 Equity Fund Categories — The Core of This Project

SEBI stands for the Securities and Exchange Board of India. Headquartered in Mumbai, it is the supreme regulatory body that oversees the Indian securities and commodity markets. SEBI's primary mission is to protect investors, enforce fair trading practices, and prevent market manipulation like insider trading. SEBI has legally defined a fixed set of equity mutual fund categories. Every actively managed equity fund must declare which single category it belongs to, and is then bound by law to keep its portfolio within that category's allocation rules.

This is the single most important concept in the entire project. Style drift is precisely defined as: a fund's actual holdings deviating from the allocation rule of its declared category.

What "market cap" means

"Market cap" is just a way to describe the **size of a company**.

- **Large cap** = big, well-known companies.
- **Mid cap** = medium-sized companies.
- **Small cap** = smaller companies that may grow fast, but can also be riskier

The market-cap based categories

Category	Mandated Minimum Allocation
Large Cap Fund	Minimum 80% in large-cap stocks (raised from 65% by SEBI in Feb 2026)
Mid Cap Fund	Minimum 65% in mid-cap stocks
Small Cap Fund	Minimum 65% in small-cap stocks
Large & Mid Cap Fund	Minimum 35% each in large-cap and mid-cap stocks
Multi Cap Fund	Minimum 25% each in large-cap, mid-cap, and small-cap stocks

Category	Mandated Minimum Allocation
Flexi Cap Fund	Minimum 65% in equity, with no market-cap restriction — manager has full freedom

2026 Regulatory Update — Know This Before You Start

In February 2026, SEBI raised the minimum equity exposure requirement for several categories, most notably Large Cap Funds (65% → 80%). It also introduced stricter portfolio-overlap caps between similar categories (e.g. Value vs. Contra funds capped at 50% overlap) and a new "Life Cycle Fund" category with glide-path allocations.

Why this matters: your historical AMFI data (especially anything before March 2026) was disclosed under the OLDER thresholds. If you are working with multi-year data, you must account for this rule change as a structural break — a fund's allocation that looked "compliant" in 2024 might be measured against a different threshold than one in 2026. Document this clearly in your project notes; it shows real regulatory awareness.

Real life Analogy: **(edited example for understanding)**

Think of mutual fund rules like a restaurant buffet menu. Earlier, the buffet said a large plate should have at least 65% of one main dish, but after the rule change, it must have 80%. So if you compare old plates with new plates without noticing the updated rule, it may look like some meals are “different,” when actually the rule changed, not just the chef’s choice.

What it means in finance:
A fund that looked compliant in 2024 may be judged differently in 2026 because SEBI changed the minimum allocation rules. So old data and new data should be compared carefully, because the standard itself was updated.

Short note version:
“SEBI changed mutual fund category rules in 2026, so pre-2026 and post-2026 fund data should not be compared directly without accounting for the new thresholds.”

1.4 What Exactly Is "Style Drift"?

Style drift is when a fund manager gradually shifts a fund's actual holdings away from what the fund's stated category promises — without changing the fund's name or marketing.

A concrete example:

A fund is named and marketed as "XYZ Large Cap Fund." Per SEBI rules it must hold at least 80% in large-cap stocks. Over 18 months, the fund manager — chasing higher short-term returns — gradually increases mid and small cap exposure to 30%, breaching the mandate, while keeping the same fund name and the same "large cap, stable, low-risk" marketing language investors originally signed up for.

Why fund managers do this:

- Mid and small cap stocks often deliver higher returns in bull markets, and a manager under pressure to show strong short-term performance numbers may be tempted to chase that extra return.
- Performance-linked compensation for fund managers can create an incentive to take on more risk than the fund's mandate allows.
- In some cases, it is simply a slow, unintentional drift — the manager makes individually reasonable decisions that cumulatively pull the portfolio away from its stated style over time.

Why this harms investors:

An investor who picked a "Large Cap Fund" specifically because they wanted lower volatility and stability has unknowingly taken on small-cap-level risk. When a market downturn hits, this fund will fall much harder than the investor expected — and the investor only discovers this after the damage is done.

1.5 Resources for Layer 1

Primary text resource

- Zerodha Varsity — Module 11: Personal Finance (Mutual Funds), by Karthik Rangappa. Free, written conversationally, India-specific, actively maintained in 2026. — https://zerodha.com/varsity/module/personalfinance/
- Zerodha Varsity — "Introduction to Mutual Funds" chapter specifically (structure, AMC, fund managers) — https://zerodha.com/varsity/chapter/introduction-to-mutual-funds/
- Zerodha Varsity — "The Mutual Fund Fact Sheet" chapter (how to read scheme objectives, benchmarks, plans) — https://zerodha.com/varsity/chapter/the-mutual-fund-fact-sheet/

YouTube channels

- Varsity by Zerodha (YouTube) — video version of the Varsity modules, same author, free — https://www.youtube.com/@varsitybyzerodha
- ET Money — search "mutual fund categories explained" and "what is style drift" for short, India-specific explainer videos — https://www.youtube.com/@etmoney

Hands-on exploration (no signup needed)

- ValueResearchOnline — search any fund by name and study its live portfolio disclosure page; this is exactly the data structure you'll be working with — https://www.valueresearchonline.com

LAYER 2

Regulatory Framework & Financial Metrics

Layer 1 told you what a fund is and what drift means conceptually. Layer 2 gives you the precise, numeric machinery: exactly how stocks get classified, and exactly which financial metrics you'll calculate to measure and prove drift.

2.1 How a Stock Gets Labelled Large / Mid / Small Cap

Market Capitalisation ("market cap") is the total market value of all of a company's outstanding shares: Market Cap = Share Price × Total Number of Shares Outstanding.

SEBI does not let AMCs decide this classification themselves. Instead, AMFI (Association of Mutual Funds in India), in consultation with SEBI and the stock exchanges (NSE, BSE, MSEI), ranks every listed company by full market capitalisation and publishes an official classification list twice a year — every January and July.

Category	Official SEBI Rank Definition
Large Cap	Companies ranked 1st to 100th by full market capitalisation
Mid Cap	Companies ranked 101st to 250th by full market capitalisation
Small Cap	Companies ranked 251st and below by full market capitalisation

Critical Detail Most Beginners Miss

This list changes twice a year. A stock that is "mid cap" in January 2026 could become "large cap" by July 2026 purely because other companies' valuations changed — without the fund manager doing anything at all.

This means: when you classify a fund's holdings month by month over a multi-year period, you MUST use the AMFI classification list that was valid for that specific month — not today's list applied retroactively. Using the wrong list will create false positives, making it look like a fund drifted when it actually just held stocks that got reclassified around it.

2.2 Monthly Portfolio Disclosure — Your Raw Dataset

SEBI mandates that every mutual fund scheme publish a full portfolio disclosure every month, listing:

- Every stock (or bond) the fund currently holds
- The exact percentage of total fund assets allocated to each holding
- The sector each stock belongs to (IT, Banking, FMCG, Pharma, etc.)
- Market value and quantity of each holding

AMFI aggregates and publishes this data, free, for every registered mutual fund in India, going back many years. This is the dataset entire project is built on there is no need to scrape private financial data providers.

2.3 Portfolio Overlap

Portfolio overlap measures how similar two funds' holdings are, expressed as a percentage. If two "different" funds you own actually hold 70% of the same stocks in similar weights, you are not as diversified as you think you're paying two expense ratios for almost the same exposure.

For your project, overlap has a second, more important use: tracking a single fund's overlap with its own category's benchmark index over time. If a "Large Cap Fund" historically had 85% overlap with the Nifty 100 index but that overlap steadily drops to 50% over 12 months, that decline itself is a quantifiable drift signal.

Real-life Analogy : (edited example for understanding)

Imagine you buy two vegetable baskets from two different shops.
If both baskets contain mostly the same tomatoes, onions, and potatoes, you do not have two very different baskets — you just paid twice. That is like high portfolio overlap.
If one basket slowly starts changing into a completely different mix, that is like a fund drifting away from its benchmark.

Short note with example

“Portfolio overlap shows how much two funds own the same stocks, and it also helps track whether a fund is sticking to its benchmark. For example, if two mutual funds both hold the same 70% of stocks, buying both gives little extra diversification. Likewise, if a Large Cap Fund used to match the Nifty 100 closely but later moves far away from it, that change signals style drift.”

2026 Regulatory Note

SEBI's February 2026 circular formally caps portfolio overlap between certain fund categories at 50% (e.g. Value vs. Contra funds within the same AMC) and now requires AMCs to disclose overlap levels monthly. This validates portfolio overlap as a legitimate, regulator-endorsed metric — strengthens the credibility the project's approach.

What is this NSE?

“NSE is India’s stock exchange, and Nifty is its market index family. An index is a basket of stocks used as a benchmark. Nifty 100 tracks the top 100 companies, so investors use it to judge how large Indian companies are performing overall.”

Easy real-life example(edited example for understanding)

Think of NSE like a big shopping mall, and stocks are the shops inside it. An index like Nifty is like a top-sales list showing how the biggest and most important shops are doing together. If the top 100 shops are doing well, the Nifty 100 score goes up; if they do badly, the score goes down.nseindia+1

2.4 Sector Concentration & Top-Holding Concentration

Sector concentration is the percentage of a fund's portfolio invested in a single sector (e.g. 40% in Banking & Financial Services). A sudden, sharp jump in concentration toward one sector can indicate the manager is making a high-conviction (and higher-risk) bet that wasn't part of the fund's original diversified mandate.

Top-10 holding concentration is the percentage of the fund held in just its 10 largest positions. A fund with 80% of assets in 10 stocks behaves very differently — and carries very different risk — from a fund spreading the same AUM across 60 stocks, even if both are nominally in the same category.

2.5 Rolling Returns

A standard "5-year return" figure is calculated from one fixed start date to one fixed end date — and can be misleading, because it depends heavily on exactly when you started measuring.

Rolling returns instead calculate that same 5-year return repeatedly, shifting the start date forward by one month (or day) each time, across the entire available history. This produces a distribution of outcomes instead of one single number, revealing how consistent a fund's performance actually is.

Why this matters for drift detection:

A fund that has drifted into small-cap territory will typically show much more volatile rolling returns — great in some windows, terrible in others — compared to its stated category peers, which should behave more consistently.

2.6 Sharpe Ratio — Risk-Adjusted Return

Sharpe Ratio measures how much return a fund generates per unit of risk (volatility) it takes on. The formula, conceptually: Sharpe Ratio = (Fund's Return − Risk-Free Rate) ÷ Fund's Standard Deviation of Returns.

A higher Sharpe ratio means the fund is earning its returns more efficiently, without taking on excessive risk to get there. A lower Sharpe ratio means the fund had to take on disproportionate risk to achieve its returns.

Real-life analogy (edited example for understanding)

Imagine two delivery riders:

- **Rider A** reaches 10 houses after riding smoothly on good roads.
- **Rider B** also reaches 10 houses, but only after taking a longer, bumpier route with more effort.

Both delivered the same result, but Rider A did it more efficiently. That is like a **higher Sharpe ratio**: better return for less risk.

Easy memory trick

- **High Sharpe** = "good reward for the amount of risk."
- **Low Sharpe** = "too much risk for the reward."

Short note :

"Sharpe ratio is like fuel efficiency for investments. A higher Sharpe means the fund earned more return without taking unnecessary risk, just like a car that travels farther with less fuel."

Common risk metrics---look into this part

- **Sharpe Ratio**: compares return to total volatility, so it treats ups and downs as risk.equitiesindia+1
- **Sortino Ratio**: compares return to downside risk only, so it ignores upside swings.nipponindiaim+1
- **Beta**: measures how much a fund moves compared with its benchmark index.equitiesindia
- **Standard Deviation**: shows how much returns swing around the average.equitiesindia
- **Maximum Drawdown**: shows the biggest fall from a peak to a trough.equitiesindia
- **Calmar Ratio**: compares return to maximum drawdown.equitiesindia

Simple way to remember them

Think of a fund like a **car ride**:

- Sharpe asks, "How smooth was the ride overall?"
- Sortino asks, "How bad were the bumps going **down**?"
- Beta asks, "Did the car move like the main highway traffic?"

- Max drawdown asks, “What was the worst drop in the journey?”schwab+1

When Sortino is useful

Sortino is especially helpful when you care more about avoiding losses than about ignoring upside swings. That is why it is often used when comparing funds that may be volatile but still deliver good returns.nipponindiaim+1

Short note

“Sortino ratio is like Sharpe ratio, but it only counts bad volatility. It measures how much return a fund gives for downside risk, so a higher Sortino means better reward for the losses it might expose you to.”schwab+1

Why this matters for drift detection:

If a "Large Cap Fund" has drifted into riskier mid/small cap stocks, its standard deviation of returns will rise. If its returns haven't risen proportionally, its Sharpe ratio will fall — giving you a second, independent signal (beyond raw holdings %) that something about the fund's risk character has changed.

2.7 Expense Ratio — Context, Not a Direct Feature

The expense ratio is the annual percentage fee an AMC charges to manage the fund. It is not, by itself, evidence of drift. But it adds important context: a fund that is both drifting from its mandate AND charging a high expense ratio represents a worse outcome for investors than drift alone — they're paying a premium fee for a risk profile they never signed up for.

Easy real-life analogy (edited example for understanding)

Think of it like booking a **premium taxi** to go to the airport.
If the taxi takes the wrong route, that is bad.
If it also charges you a premium fare, then you are paying extra for a bad ride.
So the wrong route is the main problem, and the high fare makes the problem even worse.investor.

They are saying:

- **Expense ratio alone** does not mean a fund is bad.
- **Drift alone** means the fund is not following its intended strategy.
- **Drift + high expense ratio** means you are overpaying for a fund that is not delivering the exposure you expected.investor+1

Short note for your project

“Expense ratio is the annual fee of a mutual fund. It does not prove drift, but it matters because a drifting fund with a high expense ratio is worse for investors — they are paying more for a portfolio that no longer matches the fund’s promised style.

2.8 Resources for Layer 2

Official / regulatory sources

- AMFI — Categorisation of Large, Mid & Small Cap Stocks (the official bi-annual classification list) — https://www.amfiindia.com/otherdata/categorisation-of-stocks
- SEBI — Categorization and Rationalization of Mutual Fund Schemes (official circular page) — https://www.sebi.gov.in/legal/circulars/feb-2026/categorization-and-rationalization-of-mutual-fund-schemes_99983.html

Explainer articles

- Freefincal — search "portfolio overlap mutual funds" and "rolling returns explained" (Pattu Sir's articles are direct, India-specific, and free of jargon) — https://freefincal.com
- ET Money Blog — search "mutual fund style drift India" for real, named examples of funds that were flagged — https://www.etmoney.com/learn

YouTube channels

- Freefincal / Pattu (YouTube) — deep, India-specific explainers on rolling returns, Sharpe ratio, and fund analysis — https://www.youtube.com/@freefincal
- Groww / ET Money YouTube — search "Sharpe ratio explained" and "rolling returns vs CAGR" for short visual explainers — https://www.youtube.com/@etmoney

Hands-on exploration

- ValueResearchOnline — open any fund's "Portfolio" tab to see live sector concentration and top-10 holdings exactly as you'll model them — https://www.valueresearchonline.com

LAYER 3

The ML & Statistical Concepts (Theory Only)

This layer is purely conceptual — no code, no library names, no implementation. The goal is that you understand WHY each technique fits this specific problem, so that when you eventually do build it, you're applying concepts you understand rather than copying a tutorial blindly.

3.1 Why This Is an Unsupervised Learning Problem

In supervised learning, you need labelled examples — "this fund drifted, this one didn't" — to train a model. You don't have that. No one has handed you a clean list of every fund that ever drifted, with a date attached.

This is exactly the kind of situation unsupervised learning is built for: finding structure and patterns in data without any pre-existing labels. You let the data's natural groupings reveal which funds behave like "true" large caps and which ones don't — without ever telling the algorithm in advance what the answer should be.

3.2 Clustering (K-Means) — The Core Technique

Clustering is the task of grouping data points so that points within the same group (cluster) are more similar to each other than to points in other groups — without being told beforehand what the groups should look like.

K-Means specifically works by: choosing a number of clusters (K), placing K initial "center" points, assigning every data point to its nearest center, then repeatedly recalculating each center as the average of its assigned points until the centers stop moving.

How this maps to project, conceptually:

Each fund's monthly portfolio snapshot becomes one data point, described by features like "% in large cap," "% in mid cap," "% in small cap," and sector weights. If you cluster all these monthly snapshots across many funds, you'd expect a "true large cap cluster," a "true mid cap cluster," and so on to emerge naturally. A fund whose monthly snapshots drift from sitting inside the large-cap cluster toward sitting inside the mid-cap cluster, over a sequence of months, is visually and numerically demonstrating style drift.

Why K-Means specifically (and not something else) fits well here:

- It's intuitive to explain to a non-technical interviewer: "funds that hold similar stocks in similar proportions land in the same group."
- It requires no labelled training data, which matches your real-world constraint.
- Its output (cluster assignment + distance from cluster center) is naturally interpretable as a "degree of belonging" to a category — exactly what you want to measure drift.

3.3 Anomaly Detection

Anomaly detection is the task of identifying data points that don't fit the normal pattern of the rest of the data. There are two flavours relevant to your project:

Point anomalies

A single month where a fund's holdings look very unusual compared to its peer funds in the same category — e.g. a sudden, sharp spike in small-cap exposure in one month.

Contextual / collective anomalies

A pattern across multiple consecutive months that, taken together, signals abnormal behaviour even if no single month looks extreme in isolation. This is usually the more realistic case for style drift — it tends to happen gradually, not as one dramatic month.

How distance-from-cluster-center connects clustering to anomaly detection:

Once K-Means has defined where the "true large cap" cluster sits, you can measure how far each fund's monthly data point sits from that cluster's center. A fund that consistently sits unusually far from its own declared category's cluster center — or one whose distance is steadily increasing month over month — is your anomaly signal. This single idea is what turns a plain clustering exercise into a fraud/drift detection system.

3.4 Dimensionality Reduction (PCA)

A typical fund can hold 40 to 80 different stocks. If you treat each stock's weight as a separate feature, each monthly snapshot becomes a data point with 40-80 dimensions — impossible to visualise directly and often noisy for clustering.

Principal Component Analysis (PCA) compresses many correlated features into a small number of new, uncorrelated "composite" features (called principal components) that capture as much of the original variation as possible, typically reducing 40-80 dimensions down to just 2 or 3.

Why this matters for project specifically:

After PCA, you can plot every fund's monthly snapshot on a simple 2D scatter plot. Visually, you'll be able to literally watch a dot representing "XYZ Large Cap Fund — March 2024" slowly migrate across the chart toward the mid-cap cluster over successive months. This single visual is often the most compelling thing can show in an interview or on your GitHub README — it makes an abstract numeric claim instantly understandable.

3.5 The Time Series Dimension

This project is not a single snapshot in time — it's about change over time. You are not just asking "is this fund's portfolio unusual right now?" but "is this fund's portfolio becoming progressively less like what it claims to be, month after month?"

Conceptually, this means treating each fund as having a trajectory: a sequence of cluster memberships or cluster-distances, one per month, stretched across (typically) 24 to 36 months. A flat, stable trajectory near zero distance = a well-behaved fund. A trajectory that trends consistently upward (further from its own category's cluster center over time) = a drifting fund.

A subtlety worth understanding clearly:

Short-term noise is normal and expected — funds can have one or two unusual months due to short-term cash flows, redemptions, or temporary tactical bets, then return to normal. Genuine style drift is identified by a sustained trend, not a single outlier month. Building this distinction into how you interpret your results is what separates a thoughtful project from a naive one.

3.6 Feature Engineering — What Actually Goes Into the Model

This section is conceptual only — it lists what categories of information matter, without specifying any code or formulas for computing them. From the raw monthly holdings list, the meaningful conceptual features for each fund-month snapshot are:

- Percentage of portfolio in large-cap stocks (using the AMFI classification valid for that specific month)
- Percentage of portfolio in mid-cap stocks
- Percentage of portfolio in small-cap stocks
- Sector-wise concentration (e.g. % in Banking, % in IT, % in Pharma, etc.)
- Top-10 holding concentration (how concentrated vs. diversified the fund is)
- Portfolio overlap with the fund's own declared benchmark index

Together, these form the multi-dimensional "fingerprint" of a fund in any given month — and it's this fingerprint that gets clustered, compressed via PCA, and tracked over time.

3.7 Resources for Layer 3

YouTube — conceptual / visual intuition (no coding shown, pure theory)

- StatQuest with Josh Starmer — "K-Means Clustering" — the clearest visual explanation of how K-Means works, with no code — https://www.youtube.com/c/joshstarmer
- StatQuest with Josh Starmer — "Principal Component Analysis (PCA), Step-by-Step" — https://www.youtube.com/c/joshstarmer
- StatQuest with Josh Starmer — search "Anomaly Detection" / "Outlier Detection" on the channel for the conceptual walkthrough — https://www.youtube.com/c/joshstarmer
- 3Blue1Brown — "Essence of Linear Algebra" series, useful for building real visual intuition for what PCA is doing geometrically — https://www.youtube.com/c/3blue1brown

Articles for deeper conceptual reading

- Towards Data Science — search "clustering for portfolio analysis" for practical conceptual walkthroughs connecting clustering to finance — https://towardsdatascience.com
- Towards Data Science — search "anomaly detection time series finance" for articles specifically on detecting gradual drift over time — https://towardsdatascience.com

How It All Connects — The Full Picture

Before you move to building anything, you should be able to explain this entire chain out loud, start to finish, without notes:

1. Every mutual fund declares a category (e.g. Large Cap) and is legally bound by SEBI to maintain a minimum allocation to that category (e.g. 80% in large-cap stocks).
2. Every fund must publicly disclose its complete stock holdings every month — this is your dataset, sourced from AMFI.
3. Whether a stock counts as large/mid/small cap is itself decided twice a year by AMFI's official ranking — and that ranking must be applied as-of the relevant month, not retroactively.
4. From each month's disclosure, you engineer a feature "fingerprint" for the fund: its market-cap mix, sector concentration, top-holding concentration, and benchmark overlap.
5. Clustering (K-Means) groups all fund-month fingerprints into natural categories based on actual behaviour — not declared labels.
6. PCA compresses the many features into 2-3 dimensions so the clusters and any fund's movement between them can be visualised directly.
7. By tracking a single fund's position relative to its own declared category's cluster centre across many consecutive months, you detect anomalies — and a sustained, trending anomaly (not a single noisy month) is what constitutes genuine style drift.
8. Supporting metrics — Sharpe ratio shifts, rolling return volatility, and overlap decline — give you independent, corroborating evidence beyond raw holdings percentages, making your findings more credible.

Before You Touch Any Code

Before we are ready to start building only when you can pick any random real mutual fund on ValueResearchOnline, look at its live portfolio page, and say out loud: "here's what category this claims to be, here's what its holdings actually show, and here's whether that looks consistent or not."

That single skill — reading a real fund's real data and forming a judgment — is the entire project, expressed as a sentence. Everything else is just teaching a computer to do that same judgment at scale, across hundreds of funds and many months automatically.